
ECE228 Project Proposal: Forecasting Scripps Pier Coastal Dynamics via Continuous-Time Neural ODEs

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1 Project Proposal: Track 2

1.1 Problem Background & Motivation

Accurately forecasting coastal environmental variables—such as water level, air/water temperature, and barometric pressure—is a critical engineering task for extreme weather preparedness and local climate analysis.

Traditionally, data-driven forecasting relies on discrete-time recurrent neural networks (e.g., LSTMs or GRUs). However, physical environments evolve continuously, and real-world physical monitoring sensors (like those at Scripps Pier) often suffer from irregular sampling frequencies or temporary dropouts. Discrete models struggle to naturally handle these irregular time intervals and outliers without heavy data interpolation.

Motivated by Scientific Machine Learning (SciML), this project applies **Neural Ordinary Differential Equations (Neural ODEs)** to coastal environmental forecasting. Instead of treating time-series data as discrete steps, Neural ODEs parameterize the continuous rate of change of the hidden states using a neural network. This approach aligns better with the continuous nature of physical environments, allowing us to robustly forecast future states and naturally evaluate irregularly sampled sensor data without relying on ad-hoc imputation techniques.

1.2 Contribution to UCSD Campus Life

A unique aspect of this project is its direct application to the **UCSD Scripps Institution of Oceanography (SIO)** ecosystem. By improving the reliability of local environmental forecasting, our model contributions are following:

- **Enhanced Campus Safety:** Accurate, real-time forecasting of water levels and tide surges helps the UCSD community.
- **Support for Student Athletics & Recreation:** Many UCSD students and student-athletes utilize the Black’s Beach and La Jolla Shores areas for surfing, sailing, and research. Our model provides a more robust tool for predicting water temperatures and wind speeds, directly benefiting student recreational safety.

1.3 Related Works

- **Ocean Dynamics Forecasting:** Traditional oceanographic forecasting heavily relies on computationally expensive numerical physical models (e.g., Regional Ocean Modeling System - ROMS), which solve complex fluid dynamics equations on rigid spatial grids. Recently, data-driven deep learning approaches, particularly LSTMs and Spatiotemporal Graph Neural Networks (ST-GNNs), have emerged as efficient alternatives for predicting variables like sea surface temperature and wave height directly from marine sensor data.

- **Continuous-Time Machine Learning:** Despite their computational efficiency, traditional discrete-time models struggle with the irregular sensor sampling inherent to harsh marine environments. They often rely on ad-hoc imputation that distorts underlying physical dynamics. To address this, Neural ODEs and ODE-RNNs (Chen et al., 2018; Rubanova et al., 2019) were developed to natively model continuous-time, irregularly sampled sequences. Our project adapts these Scientific Machine Learning (SciML) frameworks to ocean forecasting, bridging the gap between data-driven efficiency and continuous physical laws.

1.4 Dataset

Our primary data source is the operational environmental monitoring data provided by the National Oceanic and Atmospheric Administration (NOAA).

- **Dataset Link:** <https://tidesandcurrents.noaa.gov/stationhome.html?id=9410230>
- **Description:** We will utilize multi-year, multivariate time-series data collected from the **La Jolla, Scripps Pier, CA Station (9410230)**. The dataset includes high-resolution observations of Wind Speed, Air Temperature, Water Temperature, Barometric Pressure, and Water Levels.

1.5 Methodology

Our project will be executed through the following phases:

1. **Data Pipeline Setup:** We will programmatically fetch multi-variable records (e.g., wind speed, water level, pressure) from the Scripps Pier station using the NOAA API. Importantly, realistic sensor dropouts and missing values will be intentionally retained as irregular time steps, providing a natural testbed for evaluating continuous-time models.
2. **Continuous-Time Formulation:** Instead of updating hidden states sequentially at fixed intervals, we will formulate the environmental forecasting task as a continuous-time dynamical system. We plan to use a neural network to parameterize the continuous rate of change of the system’s hidden state. By conceptualizing the network as an ODE, we can utilize standard ODE solvers to compute future environmental states at any arbitrary continuous time point.
3. **Model Implementation & Training:** The model will be implemented using PyTorch and the `torchdiffeq` library. To train this continuous model efficiently without excessive memory overhead, we will utilize adjoint-based optimization methods, which allow us to backpropagate gradients by solving an augmented ODE backwards in time.
4. **Baseline & Robustness Evaluation:** We will implement a standard discrete-time recurrent baseline (LSTM/GRU) for comparison. Both models will be evaluated on standard forecasting accuracy metrics (MSE/MAE). Furthermore, we will explicitly test both models’ robustness on irregularly sampled sequences to demonstrate the theoretical advantages of continuous-time models in handling unpredictable sensor failure intervals.